Cart lab: change mass

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Use a controlled comparison to describe the effect of system mass on acceleration.

SUCCESS CRITERIA - I can...

1. Identify total moving mass and keep driving force fixed.
2. Compare acceleration values using a fair test.
3. Separate model data from measured evidence.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Warm-up: bring back an earlier idea

A cart accelerates uniformly from rest to 12 m/s east in 4 s. Find acceleration, average velocity and displacement.

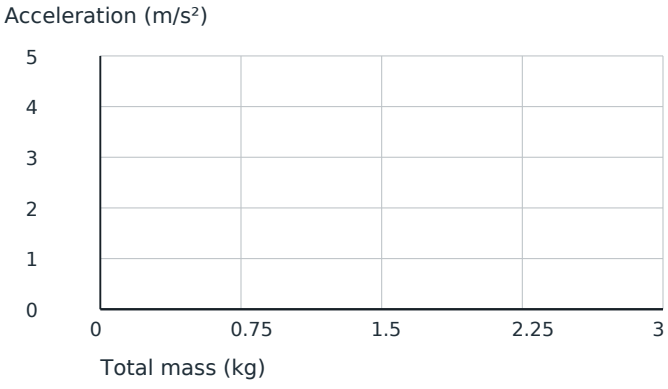
Marzano check - circle: ① ② ③

02 Worked example: explain the model

Reference: $a = F_{net}/M = m_h\, g/M$.
Ideal cart + hanger: M is all moving mass; m_h is hanging mass. Hold m_h fixed; neglect resistance and pulley inertia.

Worked reasoning: Supplied cart trials use the same net force: masses 1 and 2 kg accelerate at 4 and 2 m/s². In both cases ma=4 N; doubling mass halves acceleration.

Your turn: Plot acceleration against mass for the two trials. Label what was held constant.



Marzano check - circle: ① ② ③

03 Read the sample pattern

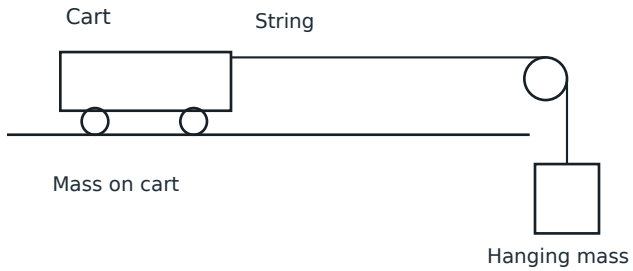
The table gives ideal cart trials with the same 4 N net force. Compare the 1 kg and 2 kg trials. What happens to acceleration when mass doubles? Support your claim with both values.

M (kg)	a (m/s²)
1	4
2	2
4	1

Marzano check - circle: ① ② ③

04 Plan a fair test

In the cart-and-hanger setup, you will change total moving mass while keeping the driving force the same. Label what you change and what you measure. Name two controls. Why would adding mass to the hanger spoil this comparison?



Marzano check - circle: ① ② ③

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Continue down each column.

05 Record evidence honestly

If equipment is available, follow the teacher's setup and record three acceleration trials at each of two total masses. Otherwise use the supplied model data in 3 and label it model data. Compare the results.

Support: My claim is ____. My two supporting values are ____.

Total M	Trial 1	Trial 2	Trial 3
____	____	____	____
____	____	____	____

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

A 4 kg sign hangs still from one vertical cord. Use g=10 N/kg. Name both forces, their sources, magnitudes and directions.
Draw the forces on the chosen object and label their sources.

Add and label force arrows.

Marzano check - circle: 1 2 3

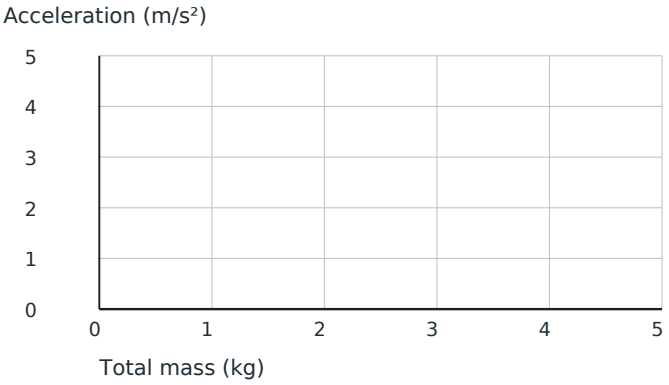
07 Exit check

At the same net force, a 1 kg system accelerates at 6 m/s². Predict acceleration for a 2 kg system. Name the condition that makes the prediction valid.

Marzano check - circle: 1 2 3

08 STRETCH: test the model

Return to the worked example in problem 2. Predict acceleration for a 4 kg cart at the same net force. Then describe a fair trial that could challenge your prediction.



Marzano check - circle: 1 2 3